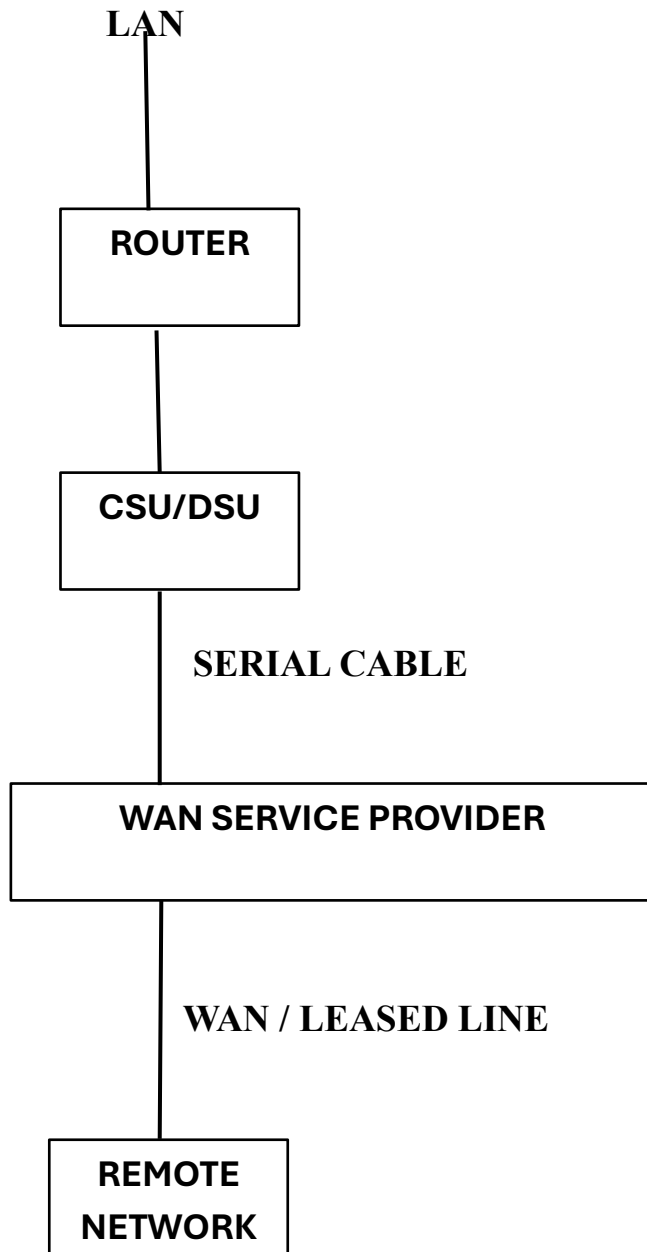


INTRODUCTION TO WAN TECHNOLOGIES

1. Create a table comparing at least three types of WAN connections (such as leased line, DSL, and MPLS) with columns for speed, cost, and typical use cases.

WAN Connection	Speed	Cost	Typical Use Case
Leased Line	High and dedicated	High	Banks, large companies, offices needing reliable dedicated connectivity
DSL	Medium	Low	Small businesses and homes for Internet/WAN access
MPLS	High and reliable	Medium–High	Connecting multiple branch offices securely
Broadband/Internet	Medium–Very High	Low–Medium	General Internet access and connecting remote offices

2. Draw a labeled diagram (on paper or using any drawing tool) showing how a CSU/DSU connects a router to a WAN service provider, and explain the role of each device in 2-3 sentences.



3. List three real-world examples of popular apps or services (like IRCTC, WhatsApp, or Zomato) that rely on WAN connections, and briefly describe how WAN enables their functionality.

1. IRCTC

IRCTC uses WAN connectivity to connect its servers, railway-related systems, payment services, and users across different locations. This allows users to search trains and make online bookings from anywhere.

2. WhatsApp

WhatsApp uses the Internet, which is a large WAN, to connect users around the world. WAN connectivity allows messages, calls, photos, and other data to travel between users and WhatsApp's servers.

3. Zomato

Zomato depends on WAN/Internet connectivity to connect customers, restaurants, delivery partners, and its servers. This allows users to browse restaurants, place orders, track deliveries, and receive updates.

4. Research online and find an image or datasheet of a physical WAN aggregator device used in enterprise networks. Paste the image or link, and write two key features of that device.

A good example is the **Cisco Catalyst 8000 Series** enterprise WAN/SD-WAN platform. Cisco describes its Catalyst 8000 platforms as supporting SD-WAN and aggregation to the cloud edge, with options for branch offices, regional hubs, and data centers.

Two Key Features

1. WAN/SD-WAN aggregation: It can aggregate WAN connections and support connectivity for enterprise branches, regional hubs, and data centers.
2. Security and intelligent routing: Cisco Catalyst SD-WAN provides built-in security and application-aware routing to select better network paths according to network conditions.